VNR VJIET

Name of the Laboratory:

Name of the Experiment:

Experiment No: **Date:**

WEEK-5:

XML Structure and Design: Book Catalog XML: Design an XML structure to represent the catalog of books in the online store. Include details such as book title, author, genre, price, and availability.

XML

- XML stands for Extensible Markup Language – it is used to store and transport data.
- It allows user-defined tags – you can create your own tags like <student>, <name>, etc.
- XML is platform independent – it works across different systems and programming languages. It follows a tree structure – data is organized in parent-child format.
- XML is both human-readable and machine-readable – easy to understand and process.

Xml code without DTD

```
<?xml version="1.0" encoding="UTF-8"?>
<student>
<name>Ashish Lukka</name>
<rollno>23071a12g5</rollno>
</student>
```

DTD

- DTD defines the structure of an XML document – it specifies which elements and attributes are allowed.
- It ensures XML validity – checks whether an XML file follows proper rules and format.
- DTD defines element relationships – parent, child, and order of elements.
- It can be written inside XML or as a separate file – internal or external DTD. DTD helps avoid data errors – by enforcing a fixed structure.

Xml code with DTD

```
<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE student [
<!ELEMENT student (name, rollno)>
<!ELEMENT name (#PCDATA)>
<!ELEMENT rollno (#PCDATA)>
]>

<student>
<name>Ashish Lukka</name>
```

23071A12G5

```
<rollno>23071a12g5</rollno>
</student>
```

Significance of DTD

1. **Defines Structure** - Specifies which elements are allowed.
2. **Validation of XML Documents** - Ensures the XML file follows predefined rules.
3. **Defines Element Order** - Controls sequence (e.g., name must come before grade).
4. **Defines Attributes** - Specifies allowed attributes and default values.
5. **Prevents Data Errors** - Reduces mistakes by enforcing structure.
6. **Ensures Data Consistency** - All XML documents follow the same forms.

Demo.xml (Without DTD)

```
<?xml version="1.0" encoding="UTF-8"?>
<catalog>

<book id="B001">
<title>Clean Code</title>
<author>Robert C. Martin</author>
<genre>Programming</genre>
<price currency="USD">35.99</price>
<availability>In Stock</availability>
</book>

<book id="B002">
<title>Atomic Habits</title>
<author>James Clear</author>
<genre>Self Help</genre>
<price currency="USD">18.50</price>
<availability>Out of Stock</availability>
</book>

<book id="B003">
<title>The Alchemist</title>
<author>Paulo Coelho</author>
<genre>Fiction</genre>
<price currency="USD">12.99</price>
<availability>In Stock</availability>
</book>

</catalog>
```

With DTD:

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demo.xml

```
<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE catalog [
<!ELEMENT catalog (book+)>
<!ELEMENT book (title, author, genre, price, availability)>
<!ATTLIST book id ID #REQUIRED>
<!ELEMENT title (#PCDATA)>
<!ELEMENT author (#PCDATA)>
<!ELEMENT genre (#PCDATA)>

<!ELEMENT price (#PCDATA)>
 <!ATTLIST price currency CDATA #REQUIRED>
 <!ELEMENT availability (#PCDATA)>
]>

<catalog>

 <book id="B001">
 <title>Clean Code</title>
 <author>Robert C. Martin</author>
 <genre>Programming</genre>
 <price currency="USD">35.99</price>
 <availability>In Stock</availability>
 </book>

 <book id="B002">
 <title>Atomic Habits</title>
 <author>James Clear</author>
 <genre>Self Help</genre>
 <price currency="USD">18.50</price>
 <availability>Out of Stock</availability>
 </book>

 <book id="B003">
 <title>The Alchemist</title>
 <author>Paulo Coelho</author>
 <genre>Fiction</genre>
 <price currency="USD">12.99</price>
 <availability>In Stock</availability>
```

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```
</book>

</catalog>
```

OUTPUT:

This XML file does not appear to have any style information associated with it. The document tree is shown below.

```
▼<catalog>
  ▼<book id="B001">
      <title>Clean Code</title>
      <author>Robert C. Martin</author>
      <genre>Programming</genre>
      <price currency="USD">35.99</price>
      <availability>In Stock</availability>
    </book>
  ▼<book id="B002">
      <title>Atomic Habits</title>
      <author>James Clear</author>
      <genre>Self Help</genre>
      <price currency="USD">18.50</price>
      <availability>Out of Stock</availability>
    </book>
  ▼<book id="B003">
      <title>The Alchemist</title>
      <author>Paulo Coelho</author>
      <genre>Fiction</genre>
      <price currency="USD">12.99</price>
      <availability>In Stock</availability>
    </book>
  </catalog>
```

23071A-05

VNR VJIET

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WEEK-6:

User Account XML: Create an XML structure to store user account information, including details like username, password (encrypted), email, and purchase history.

Demo.xml

```
<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE users SYSTEM "Demo.dtd">

<users>
<user id="U001">
<username>john_doe</username>
<password encryption="SHA-256">
5e884898da28047151d0e56f8dc6292773603d0d6aabbdd38...
</password>
<email>john@example.com</email>
<purchaseHistory>
<purchase>
<productName>Laptop</productName>
<productId>P101</productId>
<price currency="USD">1200</price>
<purchaseDate>2026-02-20</purchaseDate>
</purchase>
<purchase>
<productName>Wireless Mouse</productName>
<productId>P205</productId>
<price currency="USD">25</price>
<purchaseDate>2026-02-22</purchaseDate>
</purchase>
</purchaseHistory>

</user>
<user id="U002">
<username>alice_smith</username>
<password encryption="SHA-256">9c56cc51f6e40b8f9a4a1d4c6f4b2f</password>
<email>alice@example.com</email>
<purchaseHistory>
<purchase>
<productName>Smartphone</productName>
```

2307 1A12G5

```
<productId>P310</productId>
<price currency="USD">800</price>
<purchaseDate>2026-02-18</purchaseDate>
</purchase>
</purchaseHistory>
</user>
</users>
```


Demo.dtd

```
<!ELEMENT users (user+)>
<!ELEMENT user (username, password, email, purchaseHistory)>
<!ATTLIST user
        id ID #REQUIRED>
<!ELEMENT username (#PCDATA)>
<!ELEMENT password (#PCDATA)>
<!ATTLIST password
        encryption CDATA #REQUIRED>
<!ELEMENT email (#PCDATA)>
<!ELEMENT purchaseHistory (purchase*)>
<!ELEMENT purchase (productName, productId, price, purchaseDate)>
<!ELEMENT productName (#PCDATA)>
<!ELEMENT productId (#PCDATA)>
<!ELEMENT price (#PCDATA)>
<!ATTLIST price
        currency CDATA #REQUIRED>
<!ELEMENT purchaseDate (#PCDATA)>
```

This XML file does not appear to have any style information associated with it. The document tree is shown below.


```
▼<users>
  ▼<user id="U001">
     <username>john_doe</username>
     <password encryption="SHA-256"> 5e884898da28047151d0e56f8dc6292773603d0d6aabbdd38... </password>
     <email>john@example.com</email>
   ▼<purchaseHistory>
     ▼<purchase>
        <productName>Laptop</productName>
        <productId>P101</productId>
        <price currency="USD">1200</price>
        <purchaseDate>2026-02-20</purchaseDate>
      </purchase>
     ▼<purchase>
        <productName>Wireless Mouse</productName>
        <productId>P205</productId>
        <price currency="USD">25</price>
        <purchaseDate>2026-02-22</purchaseDate>
      </purchase>
     </purchaseHistory>
   </user>
  ▼<user id="U002">
     <username>alice_smith</username>
     <password encryption="SHA-256"> 9c56cc51f6e40b8f9a4a1d4c6f4b2f... </password>
     <email>alice@example.com</email>
   ▼<purchaseHistory>
     ▼<purchase>
        <productName>Smartphone</productName>
        <productId>P310</productId>
        <price currency="USD">800</price>
        <purchaseDate>2026-02-18</purchaseDate>
      </purchase>
     </purchaseHistory>
   </user>
 </users>
```